

- This exam contains 10 pages (including the front and back of this cover page) and 9 problems. Check to see if any pages are missing.
- This is the second exam and will count for 20% of your final grade. You have 105 minutes (5:15 pm - 7:00 pm) to complete this exam. You may use one index card of notes and a calculator on this exam, but no other outside sources may be consulted.

The following rules apply:

- Every problem on this exam, other than true-or-false problems, requires work of some form. An incorrect answer supported by substantially correct calculations and explanations will receive partial credit. **A correct answer that has no supporting work will not receive full credit.**
- Organize your work, in a reasonably neat and coherent way, in the space provided. Work scattered all over the page without a clear ordering will receive very little to no credit.
- There are 220 points total in this exam, as well as 1 bonus questions worth a total of 5 possible points.

Page:	3	4	5	6	7	8	9	Total
Points:	20	30	30	30	40	30	40	220
Score:								

Bonus:	
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Review Day Question Bonus:	
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1. (a) A subspace of $\mathbb{R}^n$ is any set H contained in $\mathbb{R}^n$ such that:
(1) $0 \in H$, (2) H is closed under vector addition, (3) H is closed under scalar multiplication.
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- (b) The columns of an invertible $n \times n$ matrix form a basis for $\mathbb{R}^n$.
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- (c) The value of $\text{rank}(A)$ is the dimension of $\text{Nul}(A)$.
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- (d) If $B = \{v_1, \dots, v_p\}$ is a basis for a subspace H , and if $x = c_1v_1 + \dots, c_pv_p$ for some $x \in H$, then the values of $c_1, \dots, c_p$ are the entries in the B -coordinate vector $[x]_B$.
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- (e) If $B = \{v_1, \dots, v_p\}$ is a basis for a subspace H , then any vector x in H can be written in at least two different ways as a linear combination of the vectors in B .
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- (f) If A is a triangular $n \times n$ matrix, then $\det(A)$ is the product of the entries on the main diagonal of A .
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- (g) If the columns of A are linearly dependent, then $\det(A) = 0$.
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- (h) The set of all solutions to a homogeneous linear system is the kernel of a linear transformation.
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- (i) If B is an echelon form of a matrix A , then the pivot columns of B form a basis for $\text{Col}(A)$.
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- (j) If A is an invertible matrix, then $\det(A^{-1}) = \frac{1}{\det(A)}$.
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2. 30 points In this question, you are given a matrix A and an echelon form of A . Use this information to find a basis for $\text{Col}(A)$ and $\text{Nul}(A)$.

$$A = \begin{bmatrix} 4 & 5 & 9 & -2 \\ 6 & 5 & 1 & 12 \\ 3 & 4 & 8 & -3 \end{bmatrix} \xrightarrow{\text{Row Reduce}} \begin{bmatrix} 1 & 2 & 6 & -5 \\ 0 & 1 & 5 & -6 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

3. 30 points Use determinants to decide if the set of vectors $\left\{ \begin{bmatrix} 4 \\ 6 \\ 2 \end{bmatrix}, \begin{bmatrix} -7 \\ 0 \\ 7 \end{bmatrix}, \begin{bmatrix} -3 \\ -5 \\ -2 \end{bmatrix} \right\}$ is linearly independent.

4. 30 points Consider the polynomials $p_1(t) = 1 + t^2$ and $p_2(t) = 1 - t^2$. Is $\{p_1, p_2\}$ a linearly independent set in $\mathbb{P}_3$? Why or why not?

5. 20 points The set $B = \left\{ \begin{bmatrix} 2 \\ -9 \end{bmatrix}, \begin{bmatrix} 1 \\ 8 \end{bmatrix} \right\}$ is a basis for $\mathbb{R}^2$. Find the change-of-coordinates matrix from the basis B to the standard basis of $\mathbb{R}^2$.

6. 20 points Let $p_1(t) = 1 + t^2$, $p_2(t) = t - 3t^2$, and $p_3(t) = 1 + t - 3t^2$. The set $B = \{p_1, p_2, p_3\}$ forms a basis for $\mathbb{P}_2$. Find the polynomial $q \in \mathbb{P}_2$ given that $[q]_B = \begin{bmatrix} -1 \\ 1 \\ 2 \end{bmatrix}$.

7. 30 points You may assume that $B = \{b_1, b_2\}$ and $D = \{d_1, d_2\}$ are bases for $\mathbb{R}^2$. Find the change-of-coordinates matrix from B to D and the change-of-coordinates matrix from D to B .

$$b_1 = \begin{bmatrix} 7 \\ 5 \end{bmatrix}, \quad b_2 = \begin{bmatrix} -3 \\ -1 \end{bmatrix}, \quad d_1 = \begin{bmatrix} 1 \\ -5 \end{bmatrix}, \quad d_2 = \begin{bmatrix} -2 \\ 2 \end{bmatrix}$$

8. 40 points Suppose A is an $m \times n$ matrix with real entries and suppose $b \in \mathbb{R}^m$. What must be true of $\text{rank} \left(\begin{bmatrix} A & b \end{bmatrix} \right)$ and $\text{rank}(A)$ in order for the equation $Ax = b$ to be consistent?

9. 5 points (bonus) Determine whether or not A is invertible, where A is the 6×6 matrix

$$A = \begin{bmatrix} 1 & 2 & \pi^2 & 7 & \sqrt{2} & -9 \\ 0 & -1/2 & 0 & 1/19 & \sin(9) & 12 \\ 0 & 0 & 3 & 12/\pi & \tan(9) & -12 \\ 0 & 0 & 0 & 1/3 & (9) & 12 \\ 0 & 0 & 0 & 0 & -2 & -12 \\ 0 & 0 & 0 & 0 & 0 & 6 \end{bmatrix}.$$